

Data Dictionary

Apex Manufacturing Executive Dashboard

Project Name: Apex Manufacturing Executive Dashboard
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1. Document Purpose

The Data Dictionary provides a detailed description of all tables, columns, data types, and business definitions used in the Apex Manufacturing Executive Dashboard. It serves as a reference for developers, analysts, and stakeholders to understand the underlying data model.

2. Dataset Overview

Category	Value
Industry	Manufacturing
Total Tables	10
Dashboard Pages	6
Fact Tables	5
Dimension Tables	5
Primary Tool	Microsoft Power BI

3. Table Summary

Table Name	Type	Description
Plants	Dimension	Stores manufacturing plant information.
Products	Dimension	Stores product details manufactured by the company.
Employees	Dimension	Stores employee information.
Machines	Dimension	Stores machine details and specifications.
Suppliers	Dimension	Stores supplier information.

Table Name	Type	Description
Production	Fact	Stores daily production transactions.
Quality	Fact	Stores quality inspection records.
Inventory	Fact	Stores inventory movement and stock records.
Purchase_Orders	Fact	Stores procurement and purchasing transactions.
Maintenance	Fact	Stores machine maintenance activities.

4. Table Definitions

4.1 Plants

Column	Data Type	Description
Plant_ID	Integer	Unique identifier for each manufacturing plant.
Plant_Name	Text	Name of the manufacturing plant.
Location	Text	Plant city or region.
Capacity	Whole Number	Maximum production capacity.

4.2 Products

Column	Data Type	Description
Product_ID	Integer	Unique product identifier.
Product_Name	Text	Product name.
Category	Text	Product category.
Unit_Cost	Decimal	Manufacturing cost per unit.
Selling_Price	Decimal	Selling price per unit.

4.3 Employees

Column	Data Type	Description
Employee_ID	Integer	Unique employee identifier.
Employee_Name	Text	Employee name.
Department	Text	Assigned department.

Column	Data Type	Description
Designation	Text	Job role.
Plant_ID	Integer	Associated manufacturing plant.

4.4 Machines

Column	Data Type	Description
Machine_ID	Integer	Unique machine identifier.
Machine_Name	Text	Machine name.
Machine_Type	Text	Machine category.
Plant_ID	Integer	Plant where the machine operates.
Status	Text	Operational status.

4.5 Suppliers

Column	Data Type	Description
Supplier_ID	Integer	Unique supplier identifier.
Supplier_Name	Text	Supplier name.
Location	Text	Supplier location.
Rating	Decimal	Supplier performance rating.

4.6 Production

Column	Data Type	Description
Production_ID	Integer	Unique production transaction ID.
Date	Date	Production date.
Plant_ID	Integer	Manufacturing plant reference.
Product_ID	Integer	Product reference.
Machine_ID	Integer	Machine reference.
Quantity_Produced	Whole Number	Number of units produced.
Target_Quantity	Whole Number	Planned production quantity.
Production_Cost	Decimal	Cost of production.

4.7 Quality

Column	Data Type	Description
Inspection_ID	Integer	Unique inspection record ID.
Date	Date	Inspection date.
Product_ID	Integer	Product reference.
Defective_Units	Whole Number	Number of defective units.
Inspection_Result	Text	Pass or Fail status.

4.8 Inventory

Column	Data Type	Description
Inventory_ID	Integer	Unique inventory record ID.
Product_ID	Integer	Product reference.
Available_Stock	Whole Number	Current stock quantity.
Reorder_Level	Whole Number	Minimum stock threshold.
Warehouse	Text	Storage location.

4.9 Purchase Orders

Column	Data Type	Description
PO_ID	Integer	Purchase order identifier.
Supplier_ID	Integer	Supplier reference.
Product_ID	Integer	Product reference.
Order_Date	Date	Purchase order date.
Quantity	Whole Number	Ordered quantity.
Total_Cost	Decimal	Total procurement cost.

4.10 Maintenance

Column	Data Type	Description
Maintenance_ID	Integer	Maintenance activity identifier.

Column	Data Type	Description
Machine_ID	Integer	Machine reference.
Maintenance_Date	Date	Maintenance completion date.
Maintenance_Type	Text	Preventive or corrective maintenance.
Maintenance_Cost	Decimal	Maintenance expenditure.
Downtime_Hours	Decimal	Machine downtime in hours.

5. Primary Keys

Table	Primary Key
Plants	Plant_ID
Products	Product_ID
Employees	Employee_ID
Machines	Machine_ID
Suppliers	Supplier_ID
Production	Production_ID
Quality	Inspection_ID
Inventory	Inventory_ID
Purchase_Orders	PO_ID
Maintenance	Maintenance_ID

6. Foreign Keys

Fact Table	Foreign Key	References
Production	Plant_ID	Plants
Production	Product_ID	Products
Production	Machine_ID	Machines
Employees	Plant_ID	Plants
Machines	Plant_ID	Plants
Inventory	Product_ID	Products
Purchase_Orders	Supplier_ID	Suppliers

Fact Table	Foreign Key	References
Purchase_Orders	Product_ID	Products
Maintenance	Machine_ID	Machines
Quality	Product_ID	Products

7. Data Standards

- All primary keys are unique and non-null.
 - Dates follow a standard Date format.
 - Numeric measures use Whole Number or Decimal data types.
 - Currency values are stored as Decimal.
 - Text fields use consistent naming conventions.
 - Relationships are maintained using foreign keys.
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8. Naming Convention

Object	Convention
Tables	PascalCase
Columns	Snake Case or descriptive names
Primary Keys	*_ID
Measures	Business-friendly names
KPIs	Descriptive titles

9. Conclusion

This Data Dictionary serves as the official metadata reference for the Apex Manufacturing Executive Dashboard. It provides standardized definitions for all tables, columns, keys, and business entities, ensuring consistency across data modeling, reporting, and future enhancements.